Pre-Calculus 11
Understanding and Solving Standards



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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1 Factoring and Quadratic Equations Part 1

Level 1 to Level 2

- Fully factor polynomials of different forms
  - $x^2 + bx + c, a^2 - b^2, ax^2 + bx + c$
  - GCF
  - Factor by grouping
- Solve factored quadratic equations

Level 3 to Level 4

- Solve quadratic equations by factoring
- Solve quadratic equations by quadratic formula, round final answers
- Solve quadratic equations with layered algebraic steps
- Understand when a polynomial is factor-able
  - i.e Find integer values of b such that $3x^2 + bx + 12$ is factor-able

Level 5

- Solve quadratic equations that have multiple layered algebraic steps
- Understand and work with perfect square trinomials
- Recognize quadratic forms and factor
- Model and solve quadratic application problems

Level 6

- Solve unfamiliar problems
- Answer open-ended problems with explanation

2 Exponents and Radicals

Level 1 to Level 2

- Identify if a number is rational or irrational
- Convert between radical and exponent forms
- Simplify expressions with exponent laws
- Simplify radical expressions
- Write as an entire radical
- Add and subtract simple radical expressions

Level 3 to Level 4

- Identify restrictions on variables
- Add and subtract more complex radical expressions
- Multiply radicals and simplify
- Divide radicals and simplify
- Solve simple radical equations

Level 5

- Multiply and divide more complex radical expressions
- Solve more complex radical equations
- Simplify more complex expressions using exponent laws

Level 6

- Solve unfamiliar problems
- Answer open-ended problems with explanation

3 Trigonometry

No calculators for skills in purple

Level 1 to Level 2

- Find areas of triangles
- Solve triangles using sine law
- Solve triangles using cosine law
- Sketch angles in standard position
- Identify quadrants based on given information
- State co-terminal angles
- State reference angle

Level 3 to Level 4

- **Evaluate trigonometric functions with special angles**
- **Understand trigonometric ratios in all quadrants**
  - eg. if point $P(3, -4)$ is on the terminal arm...
- **Solve simple trigonometric equations**

Level 5

- Solve more advanced trigonometric equations
  - Multi-step equations
  - Different domains
  - Non-special angles
- Model and solve application problems using Sine Law or Cosine Law
- Understand the ambiguous case of Sine Law

Level 6

- Solve unfamiliar problems
- Answer open-ended problems with explanation

4 Quadratic Functions and Equations Part 2

Level 1 to Level 2

- Identify properties of quadratics given graph
- Sketch a quadratic function in vertex form
- State properties of quadratics given function in vertex form
- Solve equations with Square Root method

Level 3 to Level 4

- State transformations using correct language
- Finding equations of parabolas given the graph or points
- Converting quadratic functions to vertex form
- Complete the square as needed
- Discriminant questions

Level 5

- Solve for exact values of more advanced quadratic equations
- Model and solve application problems involving quadratic functions

Level 6

- Solve unfamiliar problems
- Answer open-ended problems with explanation

5 Rational Expressions and Equations

Level 1 to Level 2

- State non-permissible values (NPVS)
- Simplify individual rational expressions of various forms
- Multiply and divide simple rational expressions
- Add and subtract rational expressions with like denominators

Level 3 to Level 4

- Multiply and divide rational expressions with multiple simplification steps
- Add and subtract rational expressions with unlike denominators
- Solve rational equations with simple steps

Level 5

- Simplify rational expressions with mixed operations
- Solve more complicated rational equations
- Model and solve application problems using rational equations

Level 6

- Solve unfamiliar problems
- Answer open-ended problems with explanation

6 Inequalities

Level 1 to Level 2

- Understand different notations: Interval, Set Notation, Inequality
- Solve one-variable inequalities
  - Linear
  - Quadratic (factorable)
- Solve two-variable inequalities
  - Linear
  - Quadratics in vertex form
- Understand and represent union and intersection of sets

Level 3 to Level 4

- Solve two-variable inequalities
  - Quadratics not in vertex form
- Solve one-variable inequalities
  - Multi-step
  - Simple rational forms
- Write two-variable inequalities given solution regions

Level 5

- Solve rational inequalities that require further simplifying
- Model and solve application problems using inequalities

Level 6

- Solve unfamiliar problems
- Answer open-ended problems with explanation